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The "Worst-Case Scenario": Recovery Between Repeated High-Intensity Efforts in Rugby League Match-Play

Tim J. Gabbett, BHSc (Hons), PhD*†‡§

Background: Repeated high-intensity effort (RHIE) activity is known to be a critical component of high-intensity, intermittent team sports. Despite the importance of this quality, there are few studies comparing the RHIE activity of higher- and lower-ranked teams competing in the same competition. This study characterized the distribution of recovery times between RHIE in Top 4 and Bottom 4 semi-elite rugby league players.

Hypothesis: Players from Top 4 teams will engage in a greater frequency of RHIE bouts than Bottom 4 teams, with shorter recovery periods between consecutive efforts.

Study Design: Cohort study.

Methods: A total of 104 semi-elite rugby league players from 11 teams wore global positioning system units during 28 Queensland Cup rugby league matches. Recovery between efforts was classified as <10, 11 to 20, 21 to 30, 31 to 60, and 61 to 120 seconds.

Results: The majority of RHIE bouts were performed with ≤10 seconds recovery between efforts for both Top 4 and Bottom 4 teams. Top 4 teams performed a greater proportion of RHIE bouts with ≤10 seconds recovery between efforts than Bottom 4 teams (effect size [ES], 0.27 ± 0.19), while only trivial differences (ES, 0.17 ± 0.20) were found between Top 4 and Bottom 4 teams for the proportion of RHIE with 61 to 120 seconds recovery between efforts. Wide-running forwards from Top 4 teams performed a greater proportion of RHIE bouts with ≤10 seconds recovery between efforts (ES, 0.59 ± 0.40) and a smaller proportion of RHIE bouts with 61 to 120 seconds recovery between efforts (ES, 0.62 ± 0.38) than wide-running forwards from Bottom 4 teams.

Conclusion: Rugby league players perform proportionally more RHIE bouts with ≤10 seconds recovery between efforts than any other recovery duration. In addition, Top 4 teams are more likely to complete a greater proportion of short (≤10 seconds) recovery RHIE bouts than Bottom 4 teams.

Clinical Relevance: These findings demonstrate the demanding nature of RHIE activity in rugby league. To be successful, teams should train for these demanding passages of play.

Keywords: collision sport; global positioning system; match analysis; team sports

Intermittent team sports such as hockey, soccer, and rugby league are characterized by short bursts of high-intensity activity separated by lower-intensity activity.[18] The importance of high-intensity actions such as repeated-sprints in relation to match outcome has been documented previously.[2,10,17] In field hockey and soccer, repeated-sprint ability has been recognized as an important physical quality,[12,17] and is also suggested to be critical to the outcome of a game.[17] However, in a study of elite National Rugby League match-play, it was found that repeated-sprint bouts were uncommon and occurred, on average, only

From †Gabbett Performance Solutions, Brisbane, Queensland, Australia, ‡Centre for Health Research, University of Southern Queensland, Ipswich, Queensland, Australia, and §Institute of Health and Wellbeing, Federation University, Ballarat, Victoria, Australia

*Address correspondence to Tim J. Gabbett, BHSc (Hons), PhD, Gabbett Performance Solutions, Brisbane, Queensland, 4011, Australia (email: tim@gabbettperformance.com.au) (Twitter: @TimGabbett).

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once per game.[7] Conversely, repeated high-intensity effort (RHIE) bouts, which also encompass tackling and physical collisions, have been shown to occur far more frequently, with players performing an average of 9 RHIE bouts per match.[7]

An RHIE bout has typically been defined as ≥3 maximal acceleration, high speed, or contact efforts with <21 seconds recovery between efforts,[14] although recently, others have also modified the definition to include ≥2 efforts.[4] Research has demonstrated that RHIE activity discriminates between elite and semi-elite rugby league players.[4,8,9] Black and Gabbett[4] found that elite rugby league players performed more RHIE bouts than semi-elite players during match-play. Furthermore, senior elite players (13.0 ± 0.8 bouts) performed significantly more RHIE bouts than junior elite players (9.7 ± 1.1 bouts).[8] These findings suggest that there is a difference between playing standards for RHIE activity. It has also been shown that there was no difference in high-speed running between winning and losing elite rugby league teams.[9] However, the RHIE demands were found to be greater for winning teams.[9] In addition, it has been documented that 56% to 70% of RHIE bouts occur within 5 minutes of a try being scored or conceded in rugby league matches.[2,11] Therefore, it is likely that RHIE ability contributes to successful rugby league match performance. Despite the importance of this quality, there are few studies comparing the RHIE activity of higher- and lower-ranked teams competing in the same competition.[3,10]

Recently, Carling et al[5] investigated the distribution of recovery times between repeated high-intensity actions in professional soccer matches and found that decrements in repeated-sprint performance occurred when efforts were undertaken with very short recovery periods (<20 and <30 seconds). In an attempt to enhance the understanding of the nature and frequency of RHIE activity, a recent rugby league study examined the performance of multiple teams in an entire competition to determine differences between higher- and lower-ranked teams.[4] When comparing elite and semi-elite players, the researchers found that elite players performed a greater frequency of RHIE bouts than semi-elite players.[4] Furthermore, elite players performed a smaller proportion of 2 effort bouts and a greater proportion of ≥4 effort bouts.[4] A limitation of current RHIE definitions is that they fail to take into account repeated high-intensity actions that could be separated by slightly greater than 21 seconds recovery. In addition, it is currently unclear if 21 seconds recovery adequately captures the most intense demands of rugby league match-play. With this in mind, the purpose of this study was to characterize the recovery time between repeated-efforts and compare these performances between Top 4 and Bottom 4 semi-elite rugby league teams.

METHODS

Experimental Approach to the Problem

This study investigated the RHIE demands of semi-elite rugby league players using a prospective cohort observational design. The RHIE demands of semi-elite players were compared in 2 ways. First, players were separated into Top 4 and Bottom 4 based on their ladder position at the completion of the season. Second, players in Top 4 and Bottom 4 teams were separated into 4 positional groups reflecting the props, wide-running forwards (second rowers and locks), adjustables (hookers, halfbacks, five-eighths, and full backs) and outside backs (centers and wingers).[10] A recovery duration distribution frequency was developed for all RHIE bouts based on the recovery duration between efforts. Effort recovery durations of ≤10, 11 to 20, 21 to 30, 31 to 60, and 61 to 120 seconds for each RHIE bout were determined for the first half, second half, and entire match.

Subjects

A total of 12 teams competing in the Queensland Cup (semi-elite) rugby league competition were invited to participate in a study of RHIE performance during match-play. All but 1 team agreed to participate in the study. The final sample included 104 male rugby league players (mean age, 24.3 ± 3.3 years). All participants were provided a clear explanation of the study, including the risks and benefits, and written consent was obtained. Experimental procedures were approved by the Institutional Review Board for Human Investigation.

Global Positioning System Analysis

Global positioning system (GPS) analysis was completed during 28 Queensland Cup matches. Movement was recorded using a minimaxX S4 GPS unit (Catapult Innovations) sampling at 10 Hz. The GPS signal provided information on speed, distance, position, and via its tri-axial accelerometer and gyroscope (sampling at 100 Hz), the unit recorded data on physical collisions and RHIE bouts. The unit was worn in a small vest, on the upper back of the player.

RHIE Activity

The definition of an RHIE bout used previously was modified in this study to ensure that physically demanding activities were reported across multiple recovery periods.[14] An RHIE bout was defined as ≥3 high-acceleration (≥2.79 m/s),[1] high-speed (≥5 m/s), or contact efforts across several recovery periods. The recovery periods used were ≤10, 11 to 20, 21 to 30, 31 to 60, and 61 to 120 seconds recovery between efforts. The 10 Hz minimaxX units have been shown to have good reliability and accuracy for measuring acceleration and sprint efforts commonly performed by team sport athletes[19] and valid measurements of tackles[13] and RHIE bouts[8] commonly observed in collision sports.

Statistical Analyses

Based upon the fact that the *P* value and traditional null-hypothesis statistics do not provide information on the size and direction of the effect and can represent an effect that is practically irrelevant,[16] magnitude-based inferential statistics were used. We chose effect sizes (ESs), confidence limits (CL), and likelihoods to provide information on the size of the

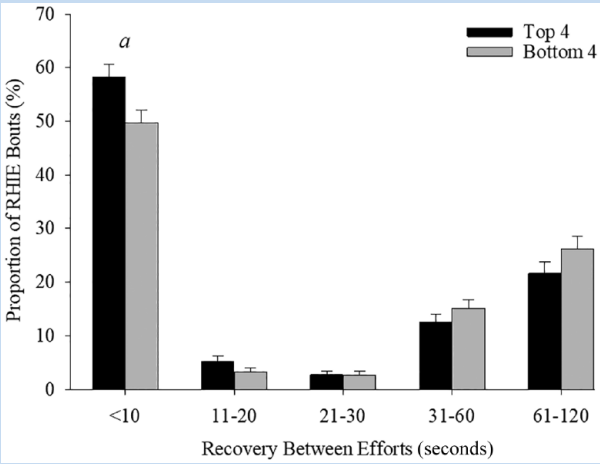

Figure 1. Proportion of RHIE bouts and recovery between efforts for Top 4 and Bottom 4 teams during semi-elite rugby league match-play. [a]Small difference between Top 4 and Bottom 4 teams. Data are reported as percentages. RHIE, repeated high-intensity effort.

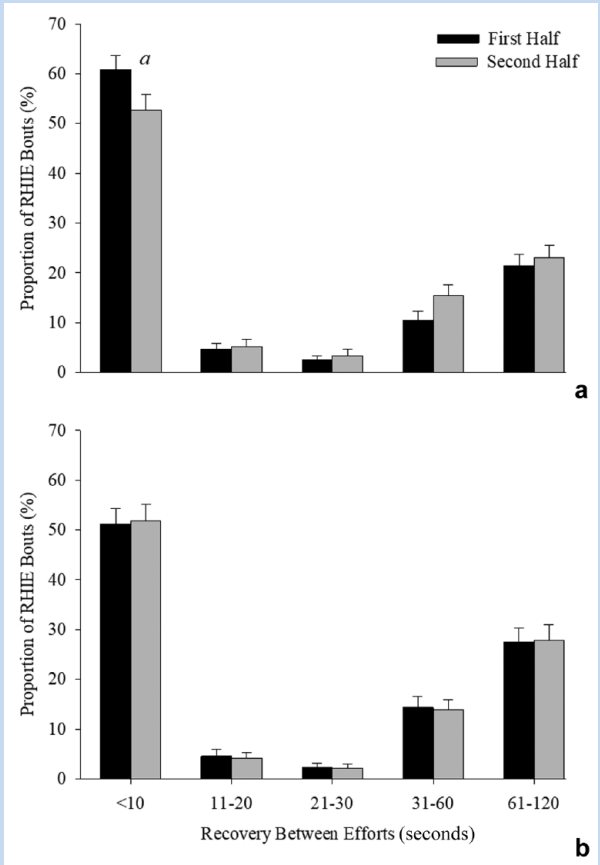

Figure 2. Proportion of RHIE bouts and recovery between efforts for the first and second half in Top 4 (a) and Bottom (b) 4 teams during semi-elite rugby league match-play. [a]Small difference between first half and second half. Data are reported as percentages. RHIE, repeated high-intensity effort.

differences, thus allowing a more practical and meaningful explanation of the results. The proportion of RHIE bouts performed per match were distributed by recovery between efforts (<10, 11–20, 21-30, 31-60, and 61-120 seconds). The proportions of different recovery durations for Top 4 and Bottom 4 teams, first and second half, and different positional groups were analyzed using Cohen's ES statistic and 90% CL.[6] ESs of <0.2, 0.2 to 0.6, 0.61 to 1.2, 1.21 to 2.0, and >2.0 were considered trivial, small, moderate, large, and very large, respectively.[14] Magnitudes of differences between the 2 groups were classified as substantially greater or lesser when there was a ≥75% likelihood of the effect being equal to or greater than the smallest worthwhile change estimated as 0.2 times between-subject standard deviation (small ES). Effects with less certainty were classified as trivial and where the ± 90% CI of the ES crossed the boundaries of ES -0.2 and 0.2, the effect was reported as unclear. The proportion of RHIE bouts for the different effort recovery durations were calculated for each player. Based on these individual percentages, the overall means were then calculated.

RESULTS

Distribution of Recovery Times Between Efforts in an RHIE Bout

The majority of recovery times in an RHIE bout were ≤10 seconds for both Top 4 and Bottom 4 teams, with Top 4 teams performing a greater proportion (17.6 ± 9.1%) of short recovery RHIE bouts than Bottom 4 teams (58.2 ± 31.1% vs 49.8 ± 30.4%; ES, 0.27 ± 0.19; 98% very likely). For bouts with very long recovery periods (61-120 seconds) between efforts, there were trivial differences (ES, 0.17 ± 0.20; 86% probable) between Bottom 4 (27.8 ± 27.7%) and Top 4 (21.4 ± 26.7%) teams. There were no meaningful differences between Top 4 and Bottom 4 teams for any of the other recovery periods (Figure 1).

Difference Between First and Second Halves

Top 4 teams performed a greater proportion of RHIE bouts with very short recovery (<10 seconds) between efforts in the first half (60.8 ± 34.6%) than the second half (52.7 ± 38.8%), although the magnitude of difference between halves was small (3.7 ± 11.3%; ES, 0.22 ± 0.22; 93% likely). There were no differences between the first and second half for Top 4 teams in any of the other recovery periods. Across all recovery periods, there were no meaningful differences in the proportion of RHIE bouts between the first and second half for Bottom 4 teams (Figure 2).

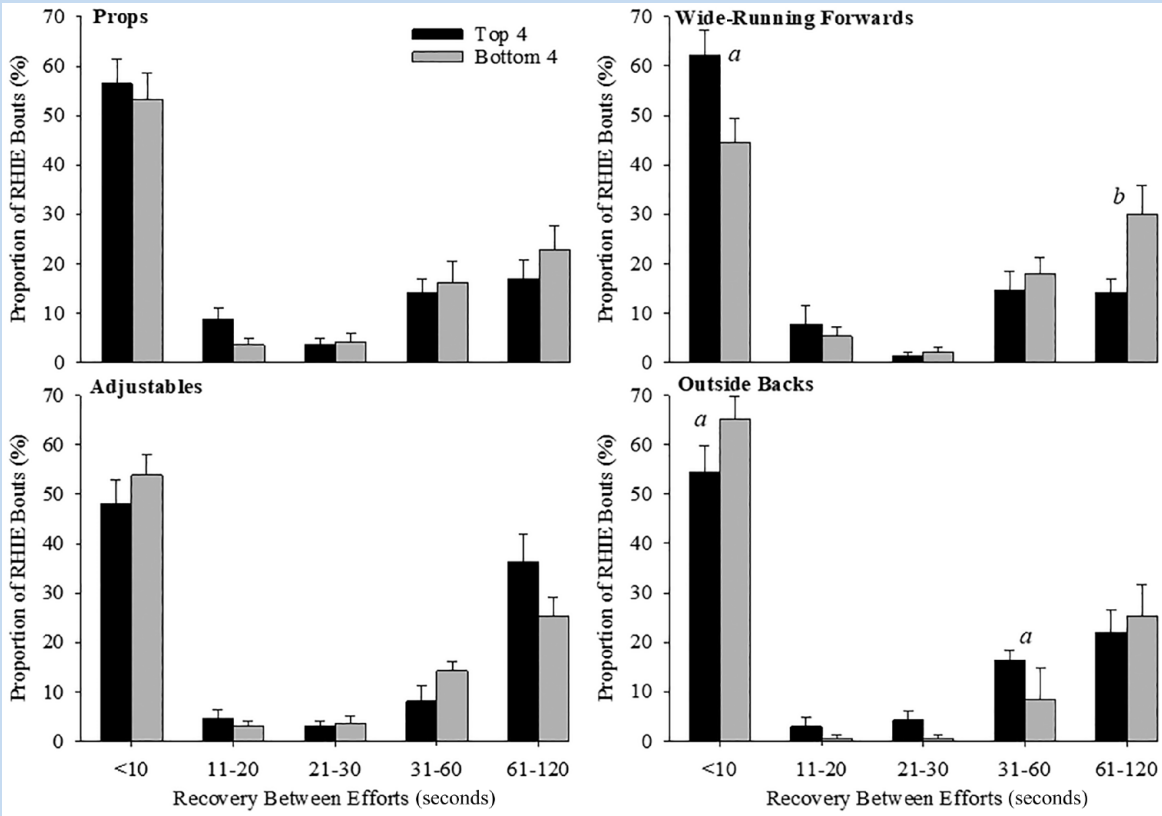

Figure 3. Proportion of RHIE bouts and recovery between efforts for different positional groups of Top 4 and Bottom 4 teams during semi-elite rugby league match-play. [a]Small difference between Top 4 and Bottom 4 teams. [b]Moderate difference between Top 4 and Bottom 4 teams. Data are reported as percentages. RHIE, repeated high-intensity effort.

Positional Differences

Props

Props from Top 4 teams had a greater proportion of 11- to 20-second recovery periods between efforts (29.4 ± 45.0%; ES, 0.49 ± 0.40; 95% likely) and a smaller proportion of 61- to 120-second recovery periods between efforts (17.6 ± 43.7%; ES, 0.25 ± 0.45; 78% likely) than props from Bottom 4 teams.

Wide-Running Forwards

Small differences (21.0 ± 19.7%; ES, 0.59 ± 0.40; 99% very likely) were found between Top 4 wide-running forwards (62.2 ± 27.4%) and Bottom 4 wide-running forwards (44.6 ± 29.6%) for the proportion of very short recovery (<10 seconds) RHIE bouts. Furthermore, there were moderate differences (53.6 ± 39.8%; ES, 0.62 ± 0.38; 99% almost certainly) between Bottom 4 wide-running forwards (29.9 ± 29.2%) and Top 4 wide-running forwards (14.2 ± 16.2%) for the proportion of very long recovery (61-120 seconds) RHIE bouts.

Adjustables

Adjustables from Top 4 teams had a smaller proportion of 31 to 60 seconds effort recovery durations (10.3 ± 36.9%; ES, 0.36 ± 0.39; 90% likely), and a greater proportion of 61 to 120 seconds effort recovery durations (11.2 ± 37.6%; ES, 0.34 ± 0.41; 89% likely) than adjustables from Bottom 4 teams.

Outside Backs

Finally, there was a small difference (9.3 ± 27.7%; ES, 0.37 ± 0.43; 91% likely) between Top 4 outside backs (51.6 ± 37.5%) and Bottom 4 outside backs (65.2 ± 33.9%) for the proportion of short recovery (<10 seconds) RHIE bouts. Outside backs from Top 4 teams had a greater proportion of 11 to 20 seconds (63.9 ± 65.0%; ES, 0.29 ± 0.37; 78% likely), 21 to 30 seconds (50.5 ± 48.0%; ES, 0.38 ± 0.36; 90% likely), and 31 to 60 seconds (15.5 ± 48.3%; ES, 0.33 ± 0.39; 89% likely) effort recovery durations than outside backs from Bottom 4 teams (Figure 3).

DISCUSSION

The results of this study demonstrate that semi-elite rugby league players perform a large proportion of RHIE bouts with ≤10 seconds recovery between efforts. Furthermore, although the difference was small, Top 4 teams were more likely to perform a greater proportion of RHIE bouts with very short recovery (<10 seconds) between efforts than Bottom 4 teams.

These findings demonstrate that rugby league RHIE activity is more demanding than previously suggested.[8,14] RHIE training should include efforts with very short recovery periods to ensure that rugby league players are adequately prepared for the most demanding passages of match-play.

While previous studies have limited the RHIE definition to a recovery duration of 21 seconds between efforts,[1,7,14,17] this study is the first to (1) investigate RHIE activity with less recovery time than the previous definition, (2) investigate the distribution of recovery times between efforts in an RHIE bout, and (3) compare the aforementioned variables between higher-ranked (Top 4) and lower-ranked (Bottom 4) teams. Both groups were more likely to perform RHIE bouts with very short recovery (<10 seconds) between efforts than any other effort recovery period. These findings suggest that RHIE activity in semi-elite rugby league match-play may be more intense than previously suggested.[8,14] Given these results, semi-elite rugby league players will likely benefit from training RHIE activity such as high-speed running, collisions, and tackling with very short recovery periods between efforts.

Small but practically meaningful differences were found between Top 4 (58.2%) and Bottom 4 (49.8%) teams for the proportion of RHIE bouts performed with <10 seconds recovery between efforts. The greater proportion of short duration recovery periods between efforts in Top 4 teams is consistent with previous studies that have found that winning teams also perform a higher frequency of RHIE bouts.[9] It appears that in comparison with Bottom 4 teams, Top 4 teams have a greater ability to recover during RHIE bouts that involve very short periods between efforts. To improve their chances of success, coaches of Bottom 4 teams should develop their players capacity to perform RHIE bouts that include recovery periods of <10 seconds between efforts. Players from Top 4 teams also performed a greater proportion of RHIE bouts with <10 s recovery between efforts in the first half of match-play compared with the second half (60.8% vs 52.7%). However, no meaningful differences were found across the recovery periods in the first and second half for Bottom 4 teams. This result is consistent with findings from others, who have shown that the pacing strategy of successful teams is based on the intention to win a match, whereas unsuccessful teams adopt a pacing strategy based on "survival."[3]

Props and wide-running forwards from Top 4 teams performed a greater proportion of RHIE bouts with <20 seconds recovery between efforts than props and wide-running forwards from Bottom 4 teams. In addition, props and wide-running forwards from Bottom 4 teams completed a greater proportion of long recovery (61-120 seconds) RHIE bouts than their Top 4 opponents. To enhance their performance in match-play, wide-running forwards from Bottom 4 teams would benefit from performing RHIE training drills designed to facilitate rapid recovery between efforts. Outside backs from Bottom 4 teams performed a greater proportion of RHIE bouts with <10 seconds recovery than outside backs from Top 4 teams. In contrast, Top 4 outside backs performed a greater proportion of RHIE bouts with 11 to 20, 21 to 30, and 31 to 60 seconds recovery than Bottom 4 outside backs. In the absence of video footage aligned with the GPS data, the differences in RHIE activity between positions from Top 4 and Bottom 4 teams is difficult to reconcile, but may reflect that the RHIE activity of some positional groups play a more important role than others in determining success in semi-elite rugby league match-play.

There are some limitations of this study that warrant discussion. First, while the majority of teams within the competition were studied, due to limited numbers of GPS devices, it was not possible to investigate the RHIE activity of every match played over the season. Second, subtle differences in tactics could explain the differences in RHIE activity between Top 4 and Bottom 4 teams; however, no attempt was made to quantify these differences in playing style. Finally, it has been shown previously that the RHIE demands of defense are greater than attack,[15] with Top 4 teams performing more RHIE bouts before conceding points, and Bottom 4 teams performing more RHIE bouts before scoring points (11). These contextual factors were not quantified in this study.

CONCLUSION

In conclusion, we found that the majority of RHIE bouts occurring in semi-elite rugby league match-play involved very short recovery (<10 seconds) between efforts. Top 4 teams were more likely to perform a greater proportion of very short recovery (<10 seconds) RHIE bouts compared with their Bottom 4 opponents. In addition, positional differences existed in RHIE activity, suggesting that position-specific training programs should be developed according to the requirements of individual positions. The ability to perform RHIE bouts with a very short recovery between efforts appears to be linked to success in semi-elite rugby league match-play.

PRACTICAL APPLICATIONS

Semi-elite rugby league players perform more RHIE bouts with <10 seconds recovery between efforts than any other recovery period. Therefore, it is recommended that strength and conditioning coaches develop programs to train players with very short recovery periods between efforts to ensure that they are adequately prepared for competition. Top 4 teams perform a greater proportion of short duration recovery periods between efforts than Bottom 4 teams. This may be due to a greater capacity to recover between efforts in an RHIE bout. To improve their chances of success, coaches of lower-ranked teams should aim to develop their player's capacity to perform RHIE bouts with <10 seconds recovery between efforts.

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